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The Rapid Adoption of Generative AI*

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Abstract

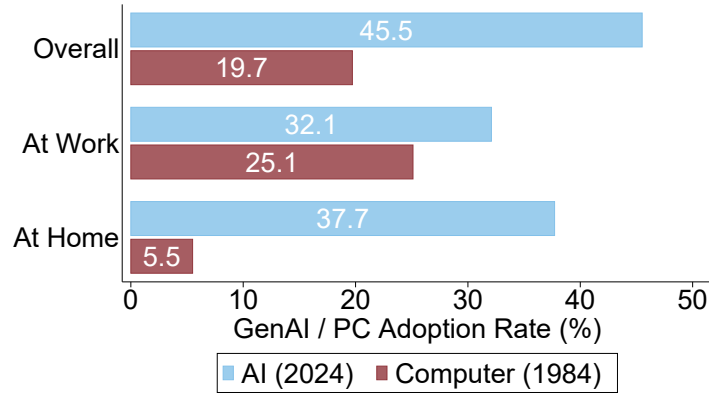
Generative artificial intelligence (AI) is a potentially important new technology, but its impact on the economy depends on the speed and intensity of adoption. This paper reports results from a series of nationally representative U.S. surveys of generative AI use at work and at home. As of late 2024, 45% of the U.S. population age 18-64 uses generative AI. Among employed respondents, 27% used generative AI for work at least once in the previous week: 10% used it every workday, and 17% on some but not all workdays. Relative to each technology's first mass-market product launch, work adoption of generative AI has been as fast as the personal computer (PC), and overall adoption has been faster than either PCs or the internet. Between 1 and 7% of all work hours are currently assisted by generative AI, and respondents report time savings equivalent to 1.4% of total work hours. Potential productivity gains vary widely by industry, and firm climate and policies play an important role in adoption patterns.

JEL Codes: J24, O33

Keywords: Generative AI, Technology Adoption, Labor Productivity

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Figure 2: Computer and Generative AI Adoption 2-3 Years After First Mass-Market Product



Notes: The figure displays usage rates for computers (PCs) and genAI in 1984 and 2024, respectively. The IBM PC was released in 1981, which we refer to as the first mass-market computer, implying that 1984 is three years after the first mass-market computer. ChatGPT was released in November 2022, which we refer to as the first mass-market genAI product, implying the fall of 2024 is two years after the first mass-market product for that technology. The data source for genAI is the August and November 2024 wave of the RPS (blue bars). The data source for computers is the 1984 Computer and Internet Use Supplement of the CPS (red bars). The sample for the Overall and At Home bars in the RPS and CPS is all individuals ages 18-64 (RPS, $N = 9726$; CPS, $N = 61,708$). The sample for the For Work bars is employed individuals ages 18-64 (RPS, $N = 6935$; CPS, $N = 89,271$).

measured using very similar questions for both technologies.) The first mass-market computer was the IBM PC, which was released in August 1981. The first mass-market genAI product was ChatGPT, which was released in November 2022.⁸ For the PC, we measure adoption in the CPS CIU for the first time in 1984, i.e. three years after the first mass-market computer. The RPS provides adoption estimates for 2024, two years after the first mass-market genAI product.

The top pair of bars in Figure 2 refers to overall adoption for the two technologies two to three years after the introduction of the first mass-market product. The bar for the RPS repeats the 45% overall adoption rate for genAI reported in Figure 1a. Overall PC adoption is roughly 20% in 1984, implying that genAI has been adopted at a faster pace than the PC. As another more recent comparison, in 1995 the internet was opened to commercial traffic after the decommissioning of NSFNet (Leiner et al., 2009).⁹ Two years later, 21% of US adults reported access to the internet according to the International Telecommunication Union, which compiles subscriber data from national regulatory bodies and service providers (Pena-Lopez et al., 2009). This is a similar pace of overall adoption to PCs, and substantially slower than the adoption of genAI.

⁸Three notable computer products had already been released in 1977 (the Apple II, the Commodore PET, and the TRS-80). Choosing earlier introduction dates would imply slower adoption rates for a given technology. The 1981 IBM PC was the first computer to sell over one million units (Abbate, 1999). We chose these dates based in part on ChatGPT’s answers to questions about the year of the first mass market product for each technology. The exact prompts and answers are available upon request.

⁹1995 was also the year of Netscape’s initial public offering and the year that AOL 3.0 was released.